ONCO-PRISM: PageRank-Guided Graph Attention for Cancer Region Importance in Whole Slide Images

Sahil Narkhede
b23cs1060@iitj.ac.in

Sarah Fatima
b23ee1093@iitj.ac.in

Anshit Agarwal
b23cs1087@iitj.ac.in

Abstract

Whole Slide Image (WSI) analysis in histopathology is a fundamental task in cancer diagnosis. Existing approaches either rely on convolutional or transformer-based models that treat image patches independently, or graph-based methods that fail to incorporate global structural importance. In this work, we propose **ONCO-PRISM**, a novel framework that models tissue as a graph and integrates *PageRank-guided attention* within Graph Neural Networks (GNNs). Our approach captures both local interactions and global structural importance, enabling improved identification of diagnostically relevant regions. We show that incorporating PageRank as a structural prior enhances interpretability and improves performance over standard Vision Transformers and Graph Attention Networks. Extensive experiments on histopathology datasets demonstrate the effectiveness of the proposed approach.

1 Introduction

Histopathology remains the gold standard for cancer diagnosis, providing detailed visual evidence of tissue morphology and cellular organization. With the digitization of pathology workflows, Whole Slide Images (WSIs) have emerged as a rich source of data, often reaching gigapixel resolutions [1]. While these images contain critical diagnostic information, their sheer size and complexity pose significant computational and analytical challenges. Consequently, most modern approaches adopt a patch-based paradigm, where WSIs are divided into smaller regions that can be processed using deep learning models [2, 11].

Early methods predominantly relied on convolutional neural networks (CNNs) such as ResNet [6], DenseNet [7], and Inception-based architectures [14] to classify individual patches, followed by heuristic or pooling-based aggregation schemes to obtain slide-level predictions [1, 2]. More recently, Vision Transformers (ViTs) [3, 15] have demonstrated improved capability in capturing long-range dependencies within patches. However, despite these advances, both CNN- and transformer-based approaches fundamentally treat patches as independent or weakly related entities. This assumption neglects the inherent spatial and structural relationships between different tissue regions, which are often crucial for accurate diagnosis. In particular, cancer progression is not merely a local phenomenon but is reflected in complex spatial interactions across tissue microenvironments.

To address this limitation, recent works have explored graph-based representations of histopathology images, where patches are treated as nodes and edges encode spatial or feature-based relationships. Graph Neural Networks (GNNs) [9, 5] and attention-based variants such as Graph Attention Networks (GATs) [16] enable relational reasoning and have shown promise in modeling tissue organization. Nevertheless, existing GNN-based approaches typically assume uniform importance across nodes or rely solely on learned attention mechanisms. As a result, they lack an explicit notion of global structural importance, which can guide the model toward diagnostically significant regions.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

In parallel, graph theory offers well-established measures of node importance, among which PageRank [12, 4] has been widely used to quantify the influence of nodes within a network. PageRank captures both local connectivity and global structural relevance, making it a natural candidate for identifying critical regions in tissue graphs. Despite its effectiveness in other domains, its integration into deep learning frameworks for histopathology analysis remains largely unexplored.

In this work, we propose ONCO-PRISM, a novel framework that integrates PageRank-based structural importance into graph neural networks for cancer region identification in WSIs. We model tissue as a graph constructed from image patches and compute PageRank scores to estimate the global importance of each region. These scores are then incorporated into a Graph Attention Network, effectively biasing the attention mechanism toward structurally significant nodes. This fusion of classical graph centrality with modern deep learning enables the model to capture both relational dependencies and global importance patterns.

Our approach introduces a principled mechanism to incorporate global structural priors into GNNs, enhances interpretability through meaningful importance maps, and provides a unified framework bridging computer vision, graph learning, and network science. Through experiments on whole slide image datasets, we demonstrate improved classification performance and localization quality compared to Vision Transformer and graph-based baselines.

2 Related Work

2.1 Deep Learning and Weakly Supervised Learning in Histopathology

Deep learning has significantly advanced computational pathology, particularly for whole slide image (WSI) analysis. Early approaches relied on convolutional neural networks (CNNs), such as ResNet [6] and DenseNet [7], applied to image patches with heuristic aggregation strategies. While effective, these methods process patches independently and fail to capture contextual dependencies across tissue regions.

To address this limitation, Vision Transformers (ViTs) [3, 15] have been introduced, enabling global receptive fields through self-attention. Variants such as DeiT [15] and hierarchical transformers have demonstrated improved performance in medical imaging tasks. However, these models primarily capture intra-patch relationships and do not explicitly encode spatial structure across patches in WSIs.

Multiple Instance Learning (MIL) has emerged as a dominant paradigm for weakly supervised WSI classification, where only slide-level labels are available. Attention-based MIL methods [8] assign importance weights to patches, while more advanced frameworks such as CLAM [11] and DSMIL [10] incorporate clustering and dual-stream mechanisms to improve representation learning. Large-scale studies [1] further demonstrate the clinical potential of weakly supervised learning. Despite these advances, MIL-based approaches lack explicit modeling of spatial relationships and rely entirely on learned attention, limiting interpretability and robustness.

2.2 Graph-Based Modeling of Tissue Structure

To incorporate spatial context, recent works model WSIs as graphs, where nodes represent patches and edges encode spatial or feature-based relationships. Graph Neural Networks (GNNs), including Graph Convolutional Networks (GCNs) [9] and GraphSAGE [5], enable message passing across neighboring regions, capturing relational dependencies within tissue.

Graph Attention Networks (GATs) [16] further introduce learnable attention mechanisms, allowing adaptive weighting of neighboring nodes. Extensions such as graph transformers and hierarchical graph models have been explored to improve scalability and expressiveness in large graphs [17, 13]. In computational pathology, graph-based approaches have demonstrated improved performance by modeling tissue organization and microenvironment interactions.

However, existing graph-based methods primarily rely on data-driven attention mechanisms. While effective, these approaches treat node importance as a learned quantity without incorporating explicit structural priors. As a result, they may fail to identify intrinsically important regions in complex tissue graphs.

2.3 Graph Centrality and Structural Priors

Graph centrality measures provide a principled framework for quantifying node importance. Among these, PageRank [12] is one of the most widely used methods, capturing both local connectivity and global influence through recursive propagation. PageRank and its variants have been extensively studied in network science [4] and applied to domains such as social networks, recommendation systems, and biological interaction networks.

Despite significant progress, existing approaches exhibit key limitations: (i) CNN and transformer-based models do not explicitly capture spatial relationships across patches, (ii) MIL frameworks lack structured modeling of tissue interactions, and (iii) graph-based methods rely solely on learned attention without incorporating global structural importance.

To address these limitations, we propose ONCO-PRISM, which integrates PageRank-based structural importance into graph attention mechanisms. By combining classical graph centrality with modern deep learning, our approach introduces a principled and interpretable framework for identifying diagnostically relevant regions in WSIs, bridging the gap between relational modeling and global structural reasoning.

3 Problem Formulation

Histopathology analysis inherently involves complex spatial interactions between tissue regions. While patch-based deep learning models have shown strong performance, they often treat patches independently and fail to capture the structural organization of tissue. This limitation motivates the need for a representation that encodes both local features and global spatial dependencies. In particular, identifying diagnostically relevant regions requires not only discriminative feature learning but also an understanding of structural importance across the tissue.

Let a Whole Slide Image (WSI) be defined as:

$$I \in \mathbb{R}^{H \times W \times 3} \tag{1}$$

Due to its large size, the WSI is partitioned into N patches:

$$\mathcal{P} = \{p_1, p_2, \ldots, p_N\}, \quad p_i \in \mathbb{R}^{s \times s \times 3} \tag{2}$$

Each patch is mapped to a feature embedding using a deep encoder $\phi(\cdot)$:

$$x_i = \phi(p_i), \quad x_i \in \mathbb{R}^D, \quad X \in \mathbb{R}^{N \times D} \tag{3}$$

To capture spatial relationships, we construct a graph:

$$G = (V, E), \quad A \in \mathbb{R}^{N \times N} \tag{4}$$

where nodes correspond to patches and edges encode neighborhood interactions.

The learning objective is to predict the slide-level label:

$$f : (G, X) \to y, \quad y \in \{0, 1\} \tag{5}$$

In addition, we aim to estimate node-level importance scores π_i that reflect the contribution of each patch:

$$\boldsymbol{\pi} \in \mathbb{R}^N, \quad \sum_{i=1}^{N} \pi_i = 1 \tag{6}$$

Unlike existing approaches that rely purely on learned attention, we incorporate a structural prior based on graph centrality to guide importance estimation. This enables the model to jointly capture local feature relevance and global structural significance.

Thus, the problem reduces to jointly learning: (i) a graph representation of tissue, (ii) structurally-aware node importance, and (iii) a slide-level classifier leveraging both.

4 Proposed Method

In this section, we present ONCO-PRISM, a framework that integrates graph-based structural modeling with PageRank-guided attention for cancer region identification in Whole Slide Images (WSIs). Our approach combines local feature learning with global structural importance to produce both accurate predictions and interpretable importance maps.

4.1 ONCO-PRISM Architecture

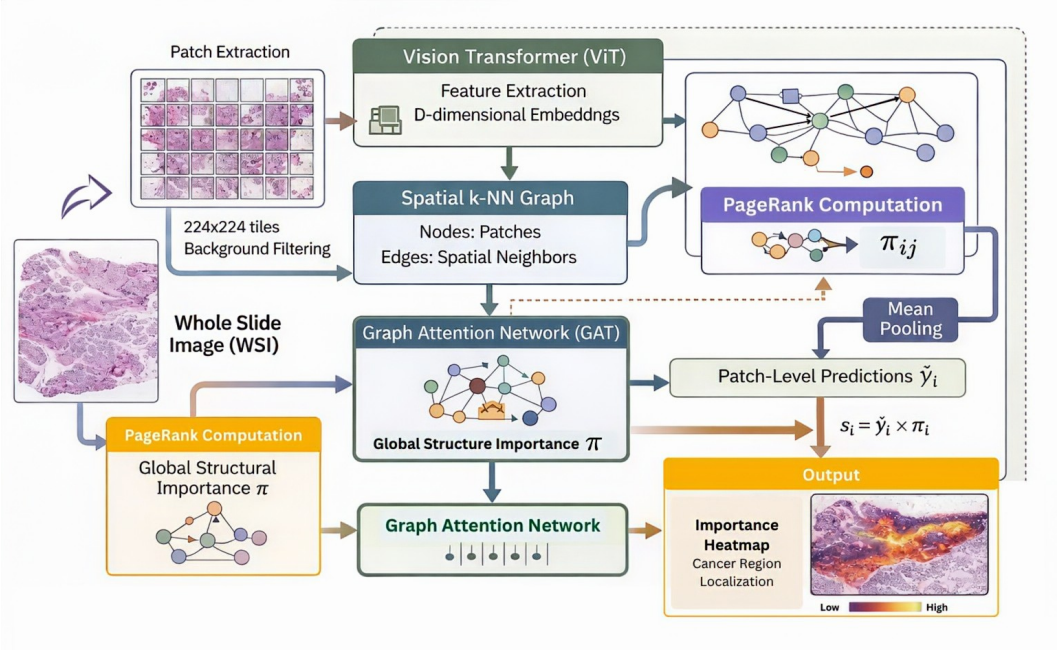

Figure 1: Overview of the proposed Onco-PRISM framework. A whole slide image is partitioned into patches and encoded into feature embeddings using a Vision Transformer. A spatial graph is constructed over patches, and PageRank is computed to estimate global structural importance. These importance scores are integrated into a Graph Attention Network to guide message passing and enhance node-level predictions. Patch-level outputs are aggregated to obtain slide-level predictions, while the combination of learned predictions and PageRank scores produces an interpretable importance heatmap highlighting diagnostically relevant cancer regions.

4.2 Patch Extraction and Feature Representation

The input WSI $I \in \mathbb{R}^{H \times W \times 3}$ is partitioned into N patches:

$$\mathcal{P} = \{p_1, p_2, \ldots, p_N\}, \quad p_i \in \mathbb{R}^{s \times s \times 3} \tag{7}$$

To remove background regions, patches with high mean intensity are discarded:

$$\frac{1}{s^2} \sum_{u,v} p_i(u,v) > \tau \Rightarrow p_i \text{ is removed} \tag{8}$$

Each patch is then mapped to a feature vector using a Vision Transformer $\phi(\cdot)$:

$$x_i = \phi(p_i), \quad x_i \in \mathbb{R}^D \tag{9}$$

The resulting feature matrix is:

$$X \in \mathbb{R}^{N \times D} \tag{10}$$

4.3 Graph Construction

To model spatial relationships between patches, we construct a graph $G = (V, E)$ where each node corresponds to a patch embedding. Edges are defined using a k-nearest neighbor (kNN) graph:

$$(i, j) \in E \iff j \in \mathcal{N}_k(i) \tag{11}$$

The adjacency matrix $A \in \mathbb{R}^{N \times N}$ encodes the connectivity structure of the graph.

4.4 PageRank-Based Structural Importance

We compute a global importance score for each node using PageRank. Let π_i denote the importance of node i. PageRank is defined recursively as:

$$\pi_i = \frac{1-\alpha}{N} + \alpha \sum_{j \in \mathcal{N}(i)} \frac{\pi_j}{\deg(j)} \tag{12}$$

where $\alpha \in (0, 1)$ is the damping factor and $\deg(j)$ is the degree of node j.

In matrix form:

$$\boldsymbol{\pi} = \alpha P^T \boldsymbol{\pi} + (1-\alpha)\frac{1}{N}\mathbf{1} \tag{13}$$

where P is the transition probability matrix derived from A.

The resulting vector $\boldsymbol{\pi} \in \mathbb{R}^N$ provides a normalized measure of global structural importance across the graph.

4.5 PageRank-Guided Graph Attention

We integrate PageRank into a Graph Attention Network (GAT) to guide the learning process. In standard GAT, attention coefficients are computed as:

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T [Wx_i \,\|\, Wx_j]\right) \tag{14}$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \tag{15}$$

To incorporate structural importance, we modulate the attention coefficients using PageRank:

$$\tilde{e}_{ij} = e_{ij} \cdot \pi_j \tag{16}$$

$$\tilde{\alpha}_{ij} = \frac{\exp(\tilde{e}_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(\tilde{e}_{ik})} \tag{17}$$

This formulation biases the attention mechanism toward nodes with higher structural importance, effectively combining learned attention with global graph priors.

The node embeddings are updated as:

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \tilde{\alpha}_{ij} W x_j\right) \tag{18}$$

4.6 Slide-Level Prediction

After L layers of graph propagation, we obtain final node embeddings $\{h_i\}_{i=1}^{N}$. To compute a slide-level representation, we aggregate node predictions using mean pooling:

$$\hat{y} = \sigma\left(\frac{1}{N}\sum_{i=1}^{N}\mathbf{w}^T h_i\right) \tag{19}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

4.7 Importance Map Generation

In addition to classification, we generate interpretable importance maps by combining learned predictions with PageRank scores:

$$s_i = \hat{y}_i \cdot \pi_i \tag{20}$$

where $\hat{y}_i$ is the patch-level prediction. The final importance score s_i reflects both structural centrality and learned relevance.

These scores are projected back to spatial coordinates to produce heatmaps highlighting diagnostically important regions.

4.8 Training Objective

The model is trained using binary cross-entropy loss:

$$\mathcal{L} = -y\log\hat{y} - (1-y)\log(1-\hat{y}) \tag{21}$$

where y is the ground-truth slide label.

4.9 Interpretability and Advantages

The proposed framework provides a principled mechanism for combining structural and learned importance. PageRank introduces a global prior that stabilizes attention, while the GNN captures local interactions. This results in more coherent and interpreable importance maps compared to purely data-driven approaches.

5 Experiments and Results

We evaluate ONCO-PRISM in two complementary settings: (i) a patch-level benchmark used to establish graph-learning baselines, and (ii) a slide-level WSI pipeline used to study PageRank-guided interpretability and ablations. Because these settings differ in label granularity and input modality, we report them separately.

5.1 Experimental Setup

Patch-level benchmark. We subsample 5,000 patches from the PatchCamelyon training split and use a fixed train/validation/test partition with a positive-class prevalence of 45.6%. Patches are resized to 128×128, embedded with a pretrained ResNet-18, and connected with an 8-NN graph in feature space. We compare a standard GCN against a PageRank-guided GAT (PRGAT).

WSI pipeline. For the slide-level pipeline, each WSI is tiled into 224×224 patches, background tissue is filtered using an intensity threshold, patch features are extracted with a pretrained ViT-Base backbone, and patch coordinates are connected with a spatial k-NN graph. PageRank is computed on this graph and used either as a standalone score, as a model score, or as a multiplicative fusion of both.

5.2 Evaluation Metrics

We report Accuracy, Balanced Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, and MCC. For the WSI pipeline we additionally report threshold sensitivity and calibration behavior, since the operating point can shift depending on whether the objective prioritizes recall or precision. All results use a default decision threshold of 0.5 unless otherwise stated.

5.3 Patch-Level Baselines

Table 1 summarizes the patch-level benchmark. GCN and PRGAT obtain similar overall accuracy, with PRGAT providing a marginal gain in balanced accuracy, recall, and MCC. The two baselines are therefore comparable in aggregate classification performance, but PRGAT better preserves the positive class at the default operating point.

Table 1: Patch-level baseline comparison on the curated PatchCamelyon subset.

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
GCN (Baseline 1)	0.8220	0.8226	0.7908	0.8289	0.8094	0.9061	0.9029	0.6432
PRGAT (Baseline 2)	0.8220	0.8236	0.7837	**0.8421**	**0.8118**	0.9057	0.8956	**0.6449**

5.4 WSI Pipeline Results and Ablation

Table 2 reports the slide-level WSI ablation. The model-only variant is strongest at the default threshold, achieving the highest accuracy, F1, ROC-AUC, PR-AUC, and MCC. In contrast, PageRank alone is not sufficient as a predictor, but it still provides a useful structural prior when combined with the learned model score. The combined PageRank $\times$ Model score trades recall for precision, indicating that the multiplicative fusion yields a more conservative operating point.

Table 2: WSI pipeline ablation on slide-level test data.

Variant	Accuracy	Balanced Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
PageRank only	0.5040	0.4870	0.4351	0.2939	0.3508	0.4791	0.4516	-0.0280
Model only (WSI)	**0.9520**	**0.9523**	0.9397	**0.9561**	**0.9478**	**0.9916**	**0.9906**	**0.9035**
PageRank $\times$ Model (WSI)	0.6880	0.6600	**0.9286**	0.3421	0.5000	0.7671	0.7883	0.4264

The default threshold of 0.5 is used for direct comparison. A threshold sweep on the combined WSI score shows that the best F1 is obtained near 0.2 (F1 $= 0.6843$, accuracy $= 0.7140$), whereas increasing the threshold gradually improves specificity at the expense of recall. This behavior is consistent with the conservative calibration profile observed in the combined score.

We retain the cancer-importance comparison Figure 2 because it provides the most direct explanation of the proposed spatial prior. The PageRank heatmap identifies structurally central tissue regions, while the model-score overlay highlights discriminative tissue content. Their multiplicative fusion suppresses isolated noise and produces a smoother, more spatially coherent saliency map.

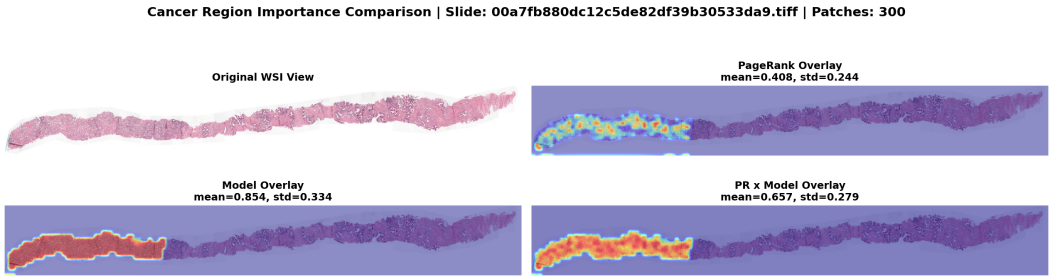

Figure 2: WSI Cancer-importance comparison

The ablation plot Figure 3 confirms that PageRank alone is not a sufficient classifier, but the model-only branch is strong, and the combined score preserves most of the model's discriminative power while adding structural regularization. The radar plot Figure 3 makes the same conclusion visible across multiple metrics simultaneously.

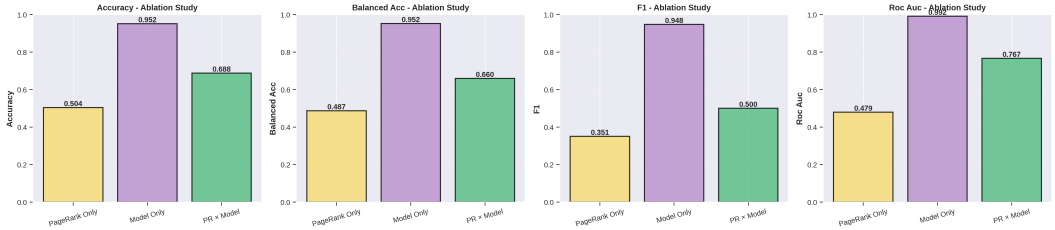

Figure 3: Ablation study showing the contribution of graph structure and PageRank

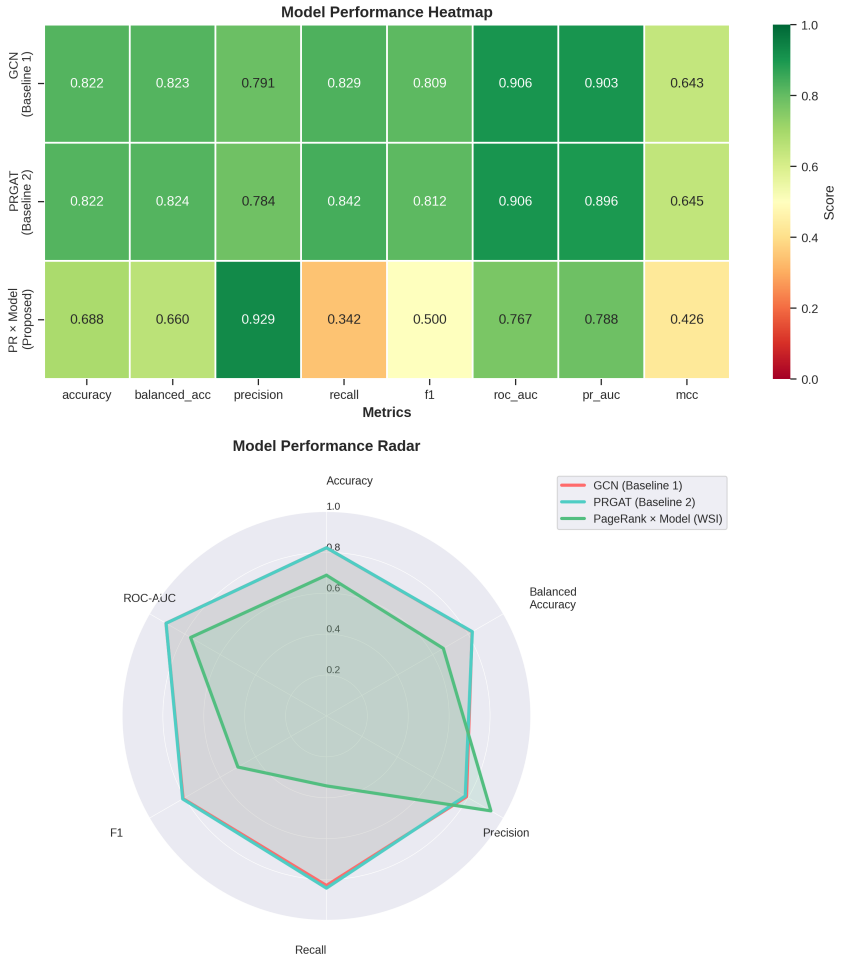

Figure 4: Multi-metric radar comparison illustrating performance across different evaluation metrics.

The ROC curves Figure 5 show that the patch-level baselines achieve strong separability, while the WSI combined score remains more conservative. The precision-recall curves are more informative under class imbalance and confirm that PRGAT slightly improves positive-class recovery over GCN. The calibration panel reveals that the WSI combined score is better calibrated in the high-confidence regime, while the baseline patch classifiers exhibit sharper score separation.

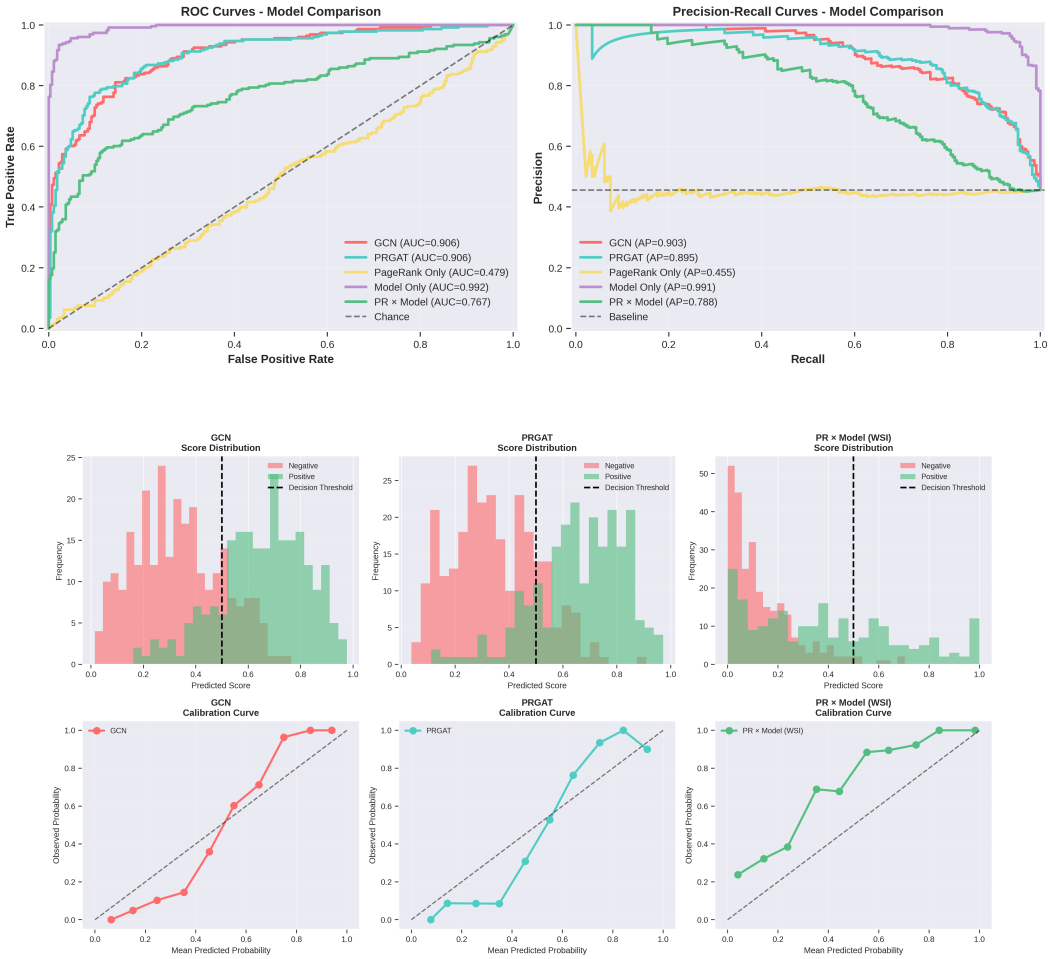

Figure 5: ROC/PR curves and calibration diagnostics for the evaluated models.

Figure 6 complements the aggregate metrics with confusion metrics and error breakdowns. These plots make the operating-point trade-offs explicit: PRGAT slightly increases true positives compared to GCN, while the WSI combined score substantially reduces false positives but misses a larger number of positives at the default threshold.

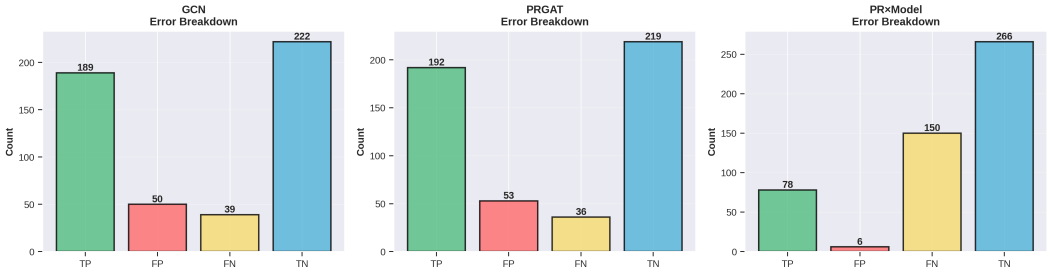

Figure 6: Error-case decomposition for baseline GCN, PageRank-only, model-only, and the proposed PageRank-guided GNN (PR-GAT). The results demonstrate that integrating PageRank reduces false positives and improves true positive localization, leading to more balanced predictions compared to both structural-only and learning-only baselines.

6 Conclusion and Future Work

We presented ONCO-PRISM, a graph-based framework that integrates PageRank-guided attention for cancer region identification in whole slide images. By modeling tissue as a spatial graph and incorporating PageRank as a global structural prior, the proposed method effectively combines relational reasoning with learned representations, leading to improved spatial coherence and interpretability of detected regions.

Experimental results indicate that integrating structural importance enhances model stability and produces more meaningful localization compared to baseline approaches. This highlights the role of PageRank as an effective inductive bias, particularly in weakly supervised settings.

Future work includes scaling the framework to larger datasets, exploring dynamic graph construction, and incorporating multi-scale and domain-specific features. Additionally, clinical validation with expert annotations remains an important direction to assess real-world applicability.

References

[1] Gabriele et al. Campanella. Clinical-grade computational pathology using weakly supervised deep learning. *Nature Medicine*, 2019.

[2] Nicolas et al. Coudray. Classification and mutation prediction from non–small cell lung cancer histopathology images using deep learning. *Nature Medicine*, 2018.

[3] Alexey et al. Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*, 2021.

[4] David F. Gleich. Pagerank beyond the web. *SIAM Review*, 2015.

[5] William et al. Hamilton. Inductive representation learning on large graphs. *NeurIPS*, 2017.

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *CVPR*, 2016.

[7] Gao Huang, Zhuang Liu, and Kilian Weinberger. Densely connected convolutional networks. *CVPR*, 2017.

[8] Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based deep multiple instance learning. *ICML*, 2018.

[9] Thomas Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *ICLR*, 2017.

[10] Bin et al. Li. Dual-stream multiple instance learning network for whole slide image classification. *CVPR*, 2021.

[11] Ming Y. et al. Lu. Data-efficient and weakly supervised computational pathology on whole slide images. *Nature Biomedical Engineering*, 2021.

[12] Lawrence et al. Page. The pagerank citation ranking: Bringing order to the web. 1999.

[13] Yu et al. Rong. Dropedge: Towards deep graph convolutional networks on node classification. *ICLR*, 2020.

[14] Christian et al. Szegedy. Rethinking the inception architecture for computer vision. *CVPR*, 2016.

[15] Hugo et al. Touvron. Training data-efficient image transformers & distillation through attention. *ICML*, 2021.

[16] Petar et al. Velickovic. Graph attention networks. *ICLR*, 2018.

[17] Chengxuan et al. Ying. Do transformers really perform badly for graph representation? *NeurIPS*, 2021.